

Karla Lopez

CAI3303C – NLP

Class Project: Tutor Module

Proposal for AI-Powered Personalized Tutor Module

1. Problem Scoping

Project Overview

The AI-Powered Tutor Module is designed to provide personalized, on-demand tutoring to college students at Miami Dade College. It aims to improve student learning outcomes by adapting to individual needs through AI-driven insights and recommendations. The project will leverage the LlamaTutor framework from Together AI as a foundation, customizing it to fit the curriculum and academic needs of Miami Dade College students.

The Tutor Module will initially support general education courses (Mathematics, English Composition, Science, and Social Sciences) and will be designed with a scalable architecture to expand into STEM, Business, and specialized fields (Nursing, IT, Engineering, etc.) in later phases.

Since we are building a personalized AI tutor for college students, a combination of LlamaTutor (chatbot) and OATutor (adaptive learning) would be ideal. LlamaTutor will handle interactive tutoring and answering queries, while OATutor will provide adaptive learning paths based on student progress.

Objectives

- Develop a personalized AI tutor that adapts to each student's academic progress.
- Provide instant feedback, recommendations, and tutoring assistance.
- Track student progress and suggest targeted learning paths.
- Integrate with Miami Dade College's course structure and learning objectives.
- Utilize Together AI's LlamaTutor as a base and enhance it with features like personalized student tracking and adaptive learning.
- Implement OATutor to provide personalized learning recommendations based on student performance.
- Ensure the Tutor Module follows an API-first design for seamless integration with Canvas and other Learning Management Systems (LMS).

Target Users

- Primary Users: College students at Miami Dade College.
- Secondary Users: Faculty, academic advisors, and administrators.

Expected Impact

- Improved student retention and understanding of complex subjects.
 - Efficient and scalable AI tutoring without requiring human intervention.
 - Support for faculty through automated insights on student learning trends.
 - Seamless integration with existing Learning Management Systems (LMS) for improved accessibility.
 - Scalable framework that allows expansion to additional courses in later phases.
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2. Data Acquisition

Required Data Sources

- Course materials (syllabi, textbooks, lecture notes).
- Student performance data (test scores, assignments, quizzes).
- Public educational resources (online learning platforms, open-source educational datasets).
- User interaction data (questions asked, response feedback, engagement metrics).
- Web API content retrieval (e.g., Bing Search, Serper API) for external academic references.

Data Collection & Privacy Considerations

- Integration with Learning Management Systems (LMS) to collect performance data.
 - Student consent & anonymization to comply with FERPA regulations.
 - Secure storage for protecting student information.
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3. Data Exploration & Preprocessing

Data Processing Steps

- Cleaning & Formatting: Handling missing data, standardizing formats.
- Text Preprocessing: Tokenization, lemmatization, and knowledge embeddings for NLP.
- Feature Engineering: Identifying key learning metrics (difficulty level, time spent, past mistakes).
- Topic Mapping: Structuring course materials for AI retrieval.

- Adaptive Learning Path Creation: Mapping students' strengths and weaknesses to personalized study plans.

Challenges & Solutions

- Incomplete student data → Use predictive models to infer missing insights.
- Course variability → Implement a flexible, modular curriculum mapping system.
- Ensuring relevance of AI responses → Use curated, college-level learning materials and real-time web retrieval for supplementary data.
- Scaling course expansion → Design a modular architecture to accommodate additional subjects seamlessly.

4. AI Modeling

AI Components

- AI Chatbot (LLM-based, LlamaTutor): Handles student queries, provides explanations, and adapts based on prior student interactions.
- Recommendation System (OATutor-based): Suggests study materials based on performance and past interactions.
- Predictive Analytics: Identifies weak areas and provides targeted study plans.
- Knowledge Graphs: Structures learning materials for efficient retrieval.
- Adaptive Tutoring Paths: Uses AI to modify content recommendations based on student progress.

Tools & Frameworks

- Programming Languages: Python
- Libraries: TensorFlow, PyTorch, Scikit-learn, OpenAI API
- Database: PostgreSQL / MongoDB
- NLP Models: GPT-4, Llama, or BERT
- Web API Integrations: Bing Search, Serper API for retrieving external educational references.
- LMS Integration: Canvas API for automated student progress tracking.

5. Evaluation

Success Metrics

- Accuracy of AI-generated responses and recommendations.
- Student engagement levels (usage frequency, feedback scores).
- Performance improvement (pre- and post-tutoring assessments).
- Faculty satisfaction with AI-generated reports and insights.

Testing Plan

- Pilot testing with a small group of students.
- A/B Testing: Comparing AI-assisted learning vs. traditional tutoring.
- User feedback collection to refine tutoring strategies.

6. Deployment & Monitoring

Deployment Plan

- Phase 1: Web-based AI tutor (Streamlit/Gradio) for early testing with general education courses.
- Phase 2: LMS integration for broader accessibility (Canvas API, Blackboard integration).
- Phase 3: Cloud deployment for scalability (Google Cloud/AWS).
- Phase 4: Continuous AI model updates based on student feedback and learning trends.
- Phase 5: Expand course coverage to STEM, Business, and specialized fields.

Monitoring & Maintenance

- User analytics tracking to measure engagement.
- Periodic AI model updates based on new course materials.
- Error handling & feedback mechanisms for continuous improvement.
- Ensuring model fairness by monitoring response accuracy and content diversity.
- Scalability adjustments to handle increased subject expansion.

Next Steps

1. Analyze LlamaTutor's architecture and identify reusable components.
2. Develop student profile system (database, authentication, adaptive learning paths).
3. Enhance chatbot with memory & personalized interactions.
4. Integrate OATutor for adaptive learning recommendations.

5. Integrate external learning resources for enriched tutoring responses.
6. Develop API-first integration for Canvas & LMS compatibility.
7. Deploy a pilot version and test with Miami Dade College students.
8. Prepare for phase-wise subject expansion post-deployment.

This proposal outlines the AI-Powered Tutor Module's role in creating an adaptive, personalized learning experience for college students, ensuring academic success through AI-driven insights and recommendations while leveraging LlamaTutor (chatbot) and OATutor (adaptive learning) as foundational frameworks. The scalable design allows for seamless expansion into new subjects and LMS integrations in the future.